

COMP6252 Deep Learning Technologies

Revision – Part 2

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Exam Format

Time: 22/05/2026 14:30

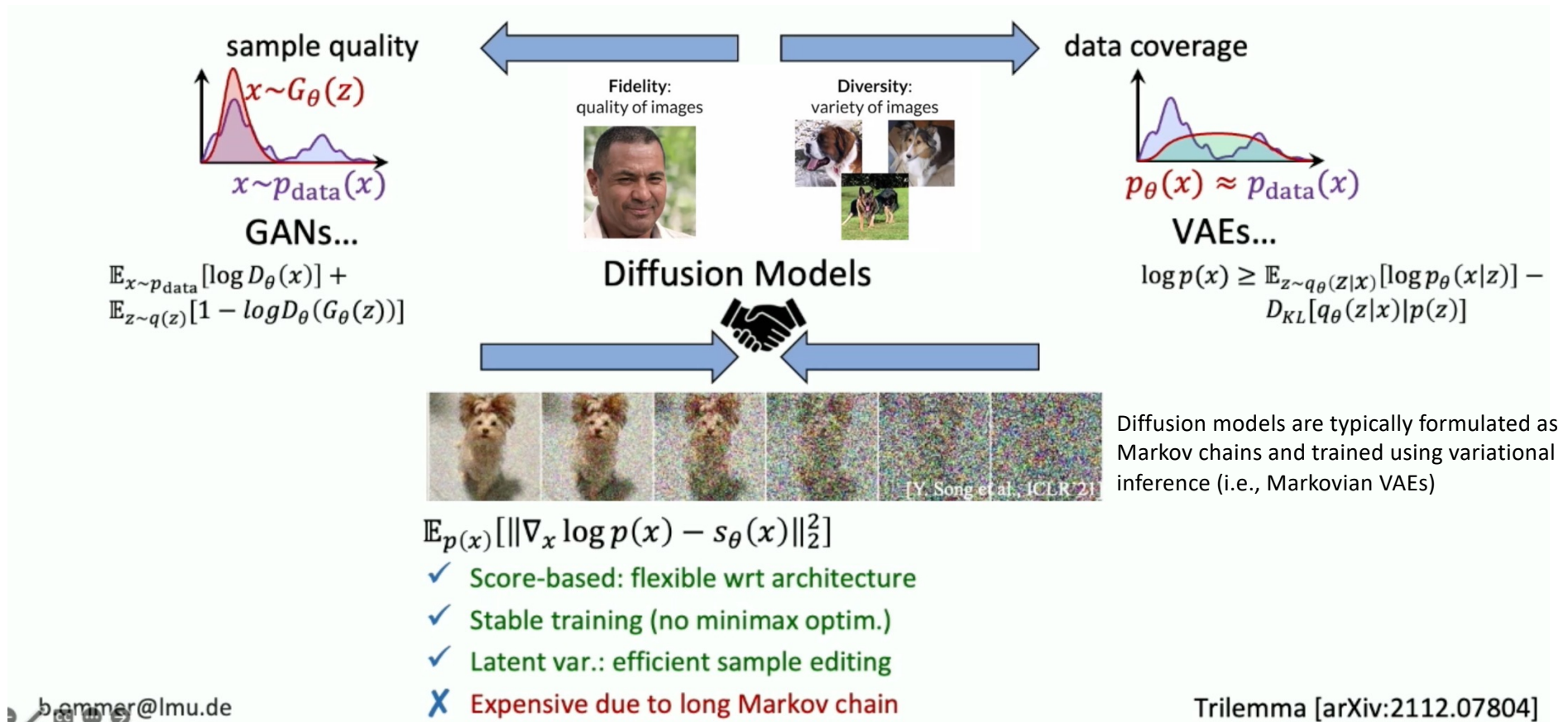
2 hours duration; **closed-book**

Please double check the
latest timetable

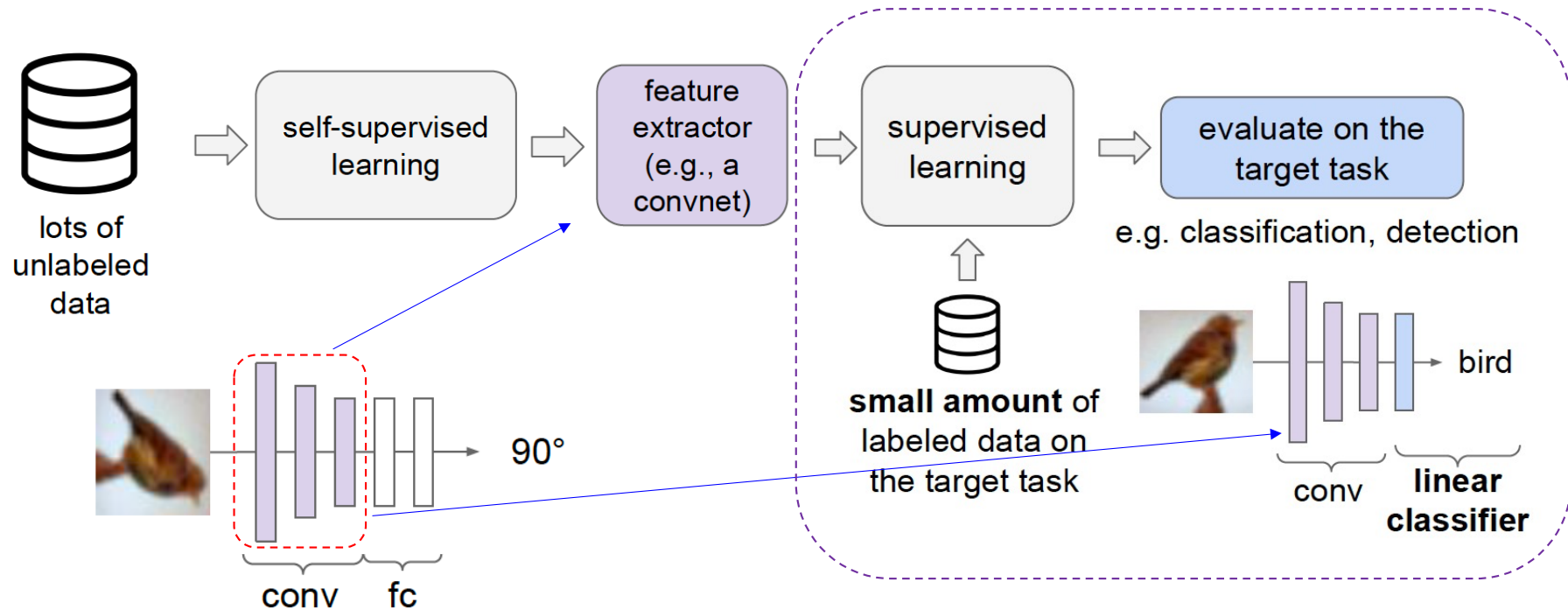
Format: Answer all the **4** questions

- Each question worth **about 25** marks (100 marks total)
- Hikmat wrote (the first) 3; Zhiwu wrote (the last) 1
 - Expect that the last question will typically cover higher level deep learning technologies & Hikmat's will cover lower level deep learning strategies.
 - but there will probably be some overlap!

Generative Modeling



Self-supervised Learning

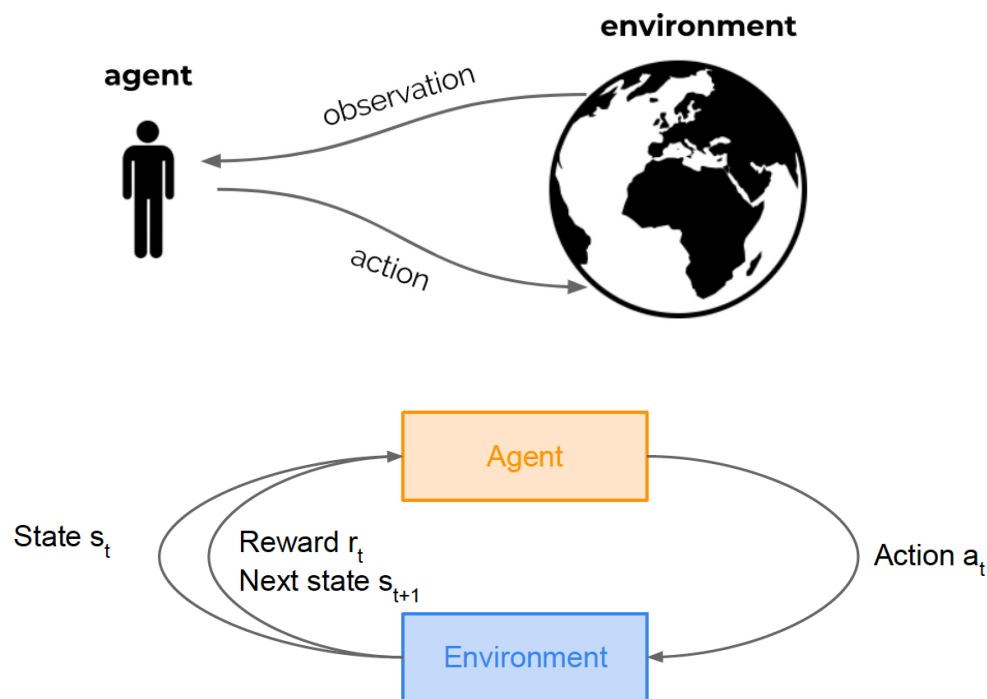


Step1: Learn general data features from pretext tasks

Step2: Transferred to downstream tasks using the learned data features from pretext tasks

Reinforcement Learning

People and animals learn by interacting with our environment



Problems involving an **agent** interacting with an **environment**, which provides numeric **reward** signals

Goal

Learn how to take **actions** in order to **maximise reward**

Characteristics

- No supervisor, only a reward signal;
- Feedback is delayed, not instantaneous
- Time really matters (sequential)
- Agent's actions affect the subsequent data it receives

The reward hypothesis

Any goal can be formalised as the outcome of maximising a cumulative reward.

Question 4.

- (a) Consider a dataset D containing millions of unlabelled images of tigers and lions. You are asked to design a deep learning method to classify images of tigers and lions.
- (i) Suppose labelling any of the images in dataset D is not allowed, design a method exploiting deep learning technologies to classify images of tigers and lions, and show the architecture of your designed method.
 - (ii) Suppose labelling a very small part of the images in dataset D is allowed, design a deep learning method to classify images of tigers and lions, and show the architecture of your designed deep learning method.

[18 marks]

- (b) ChatGPT (Chat Generative Pre-trained Transformer) is a chatbot developed by OpenAI and launched in November 2022.
- (i) Describe one main limitation of ChatGPT, and briefly provide your insights on what causes the limitation.
 - (ii) Discuss the future of Chat GPT in terms of its performance and potential regulations (from the government) being applied on it.

[7 marks]

Thanks and Good luck!